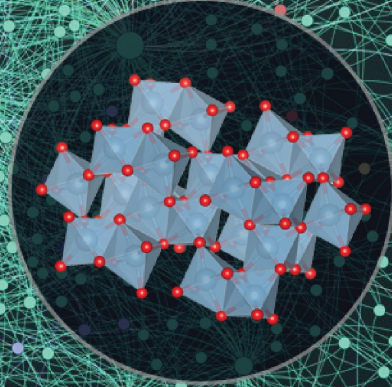
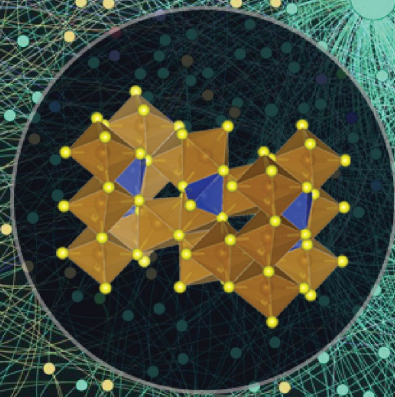
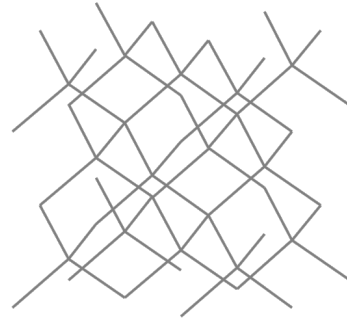


Structure-based feature vector

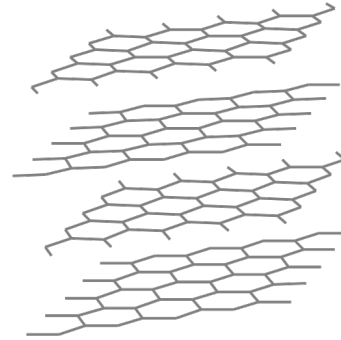




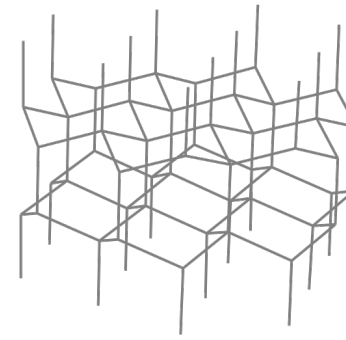
Carbon is a great example of why we can't ignore structure



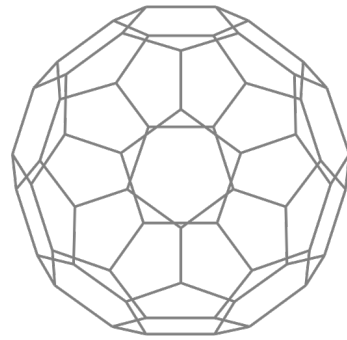
a



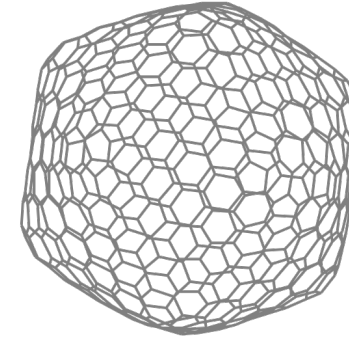
b



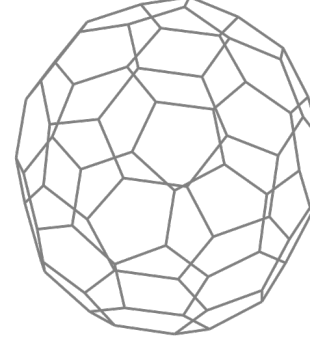
c



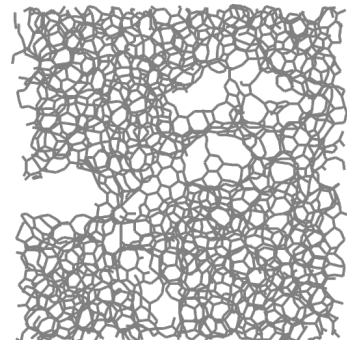
d



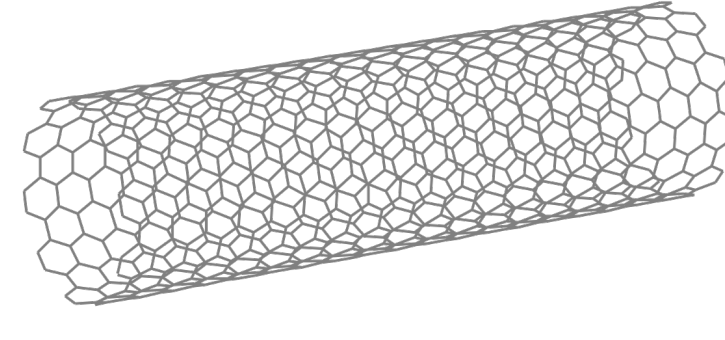
e



f



g



h

Which structure are we talking about for structural feature?

Electronic



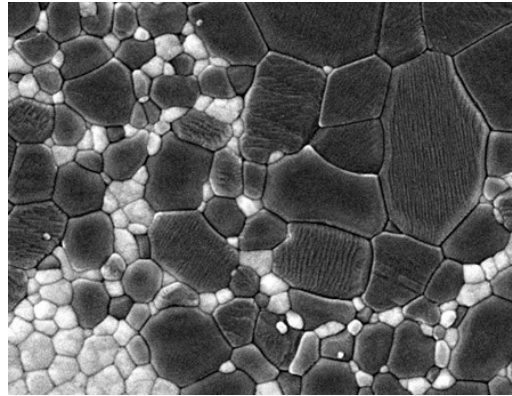
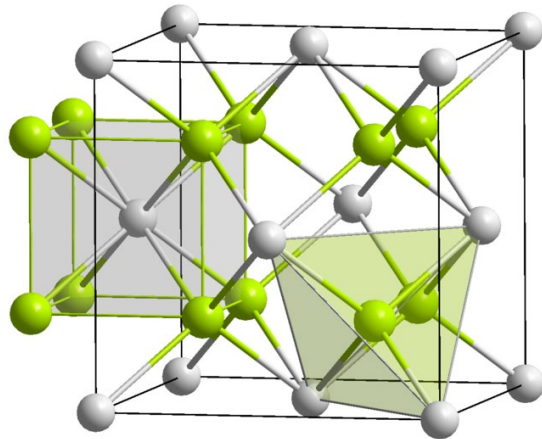
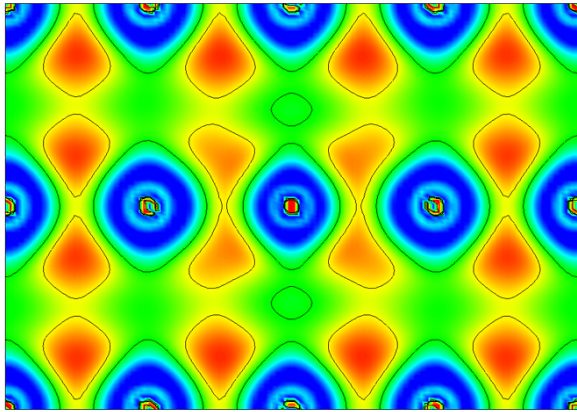
Atomic



Microstructure



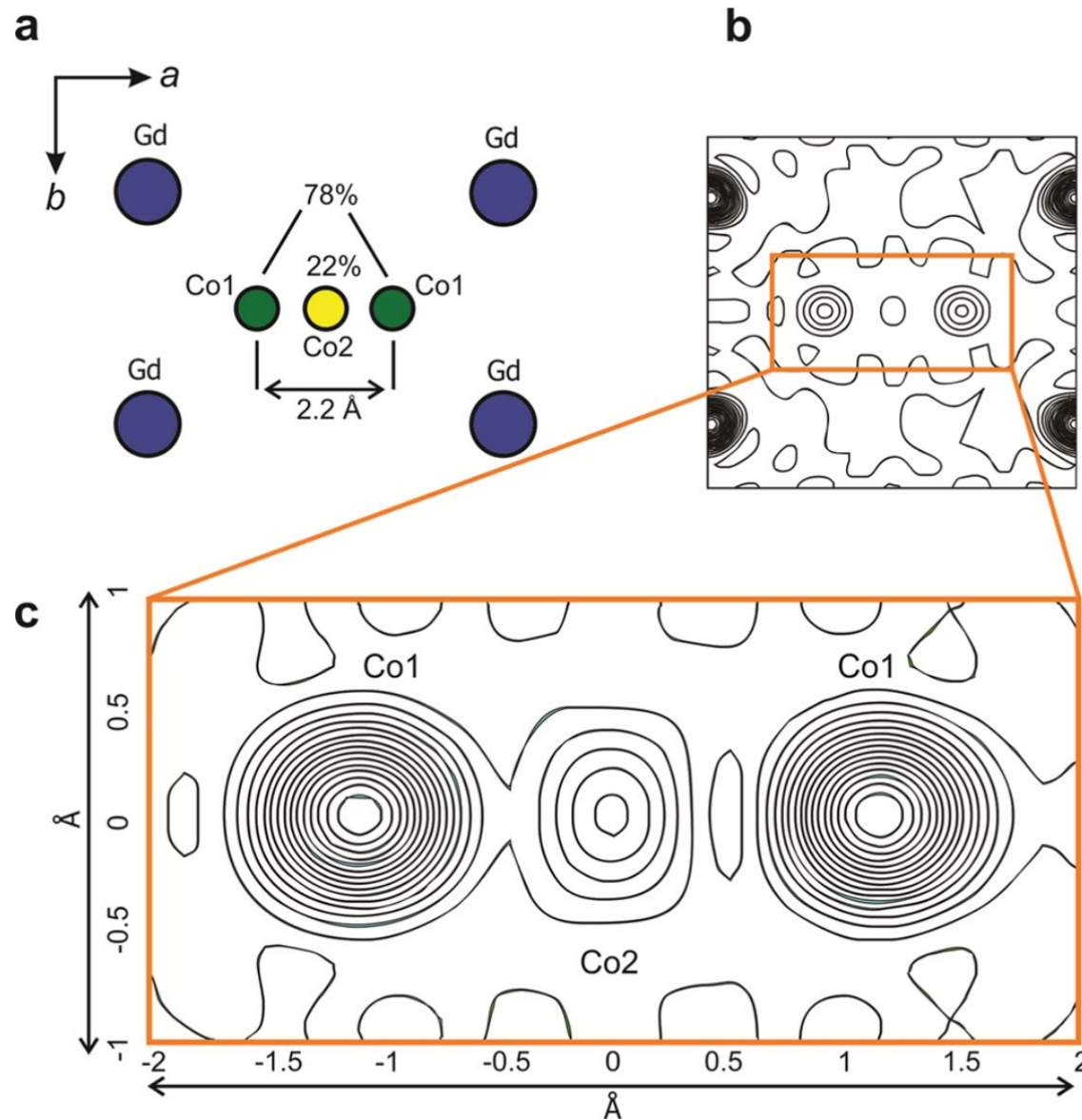
Macroscale



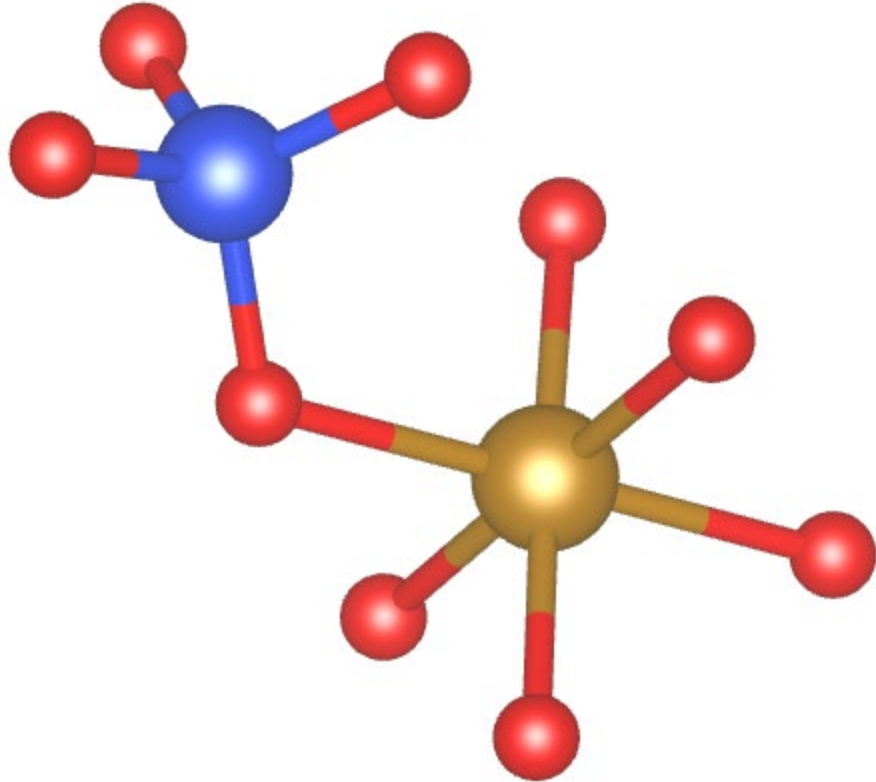
[illegible]

Atomic structure can also be approximated by electron density map

Oliyntyk et al, *Inorg Chem*, 55, 13, 6625-6633, (2016)



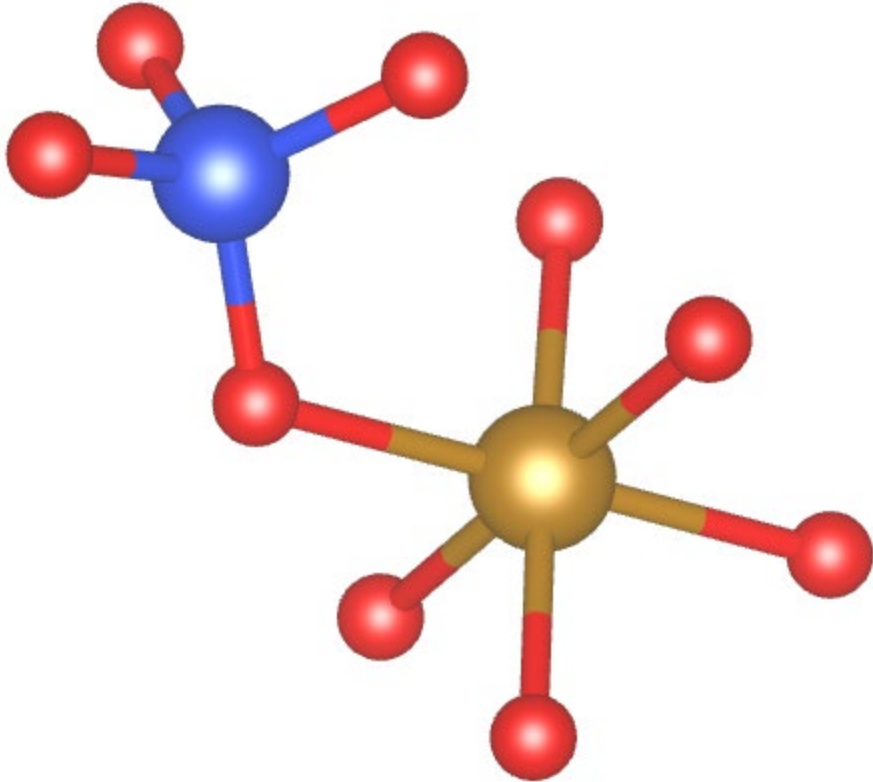
Crystal structure has different features at different scales



Bonds

- Type
- Elements involved
- Length
- Distribution of different bonds

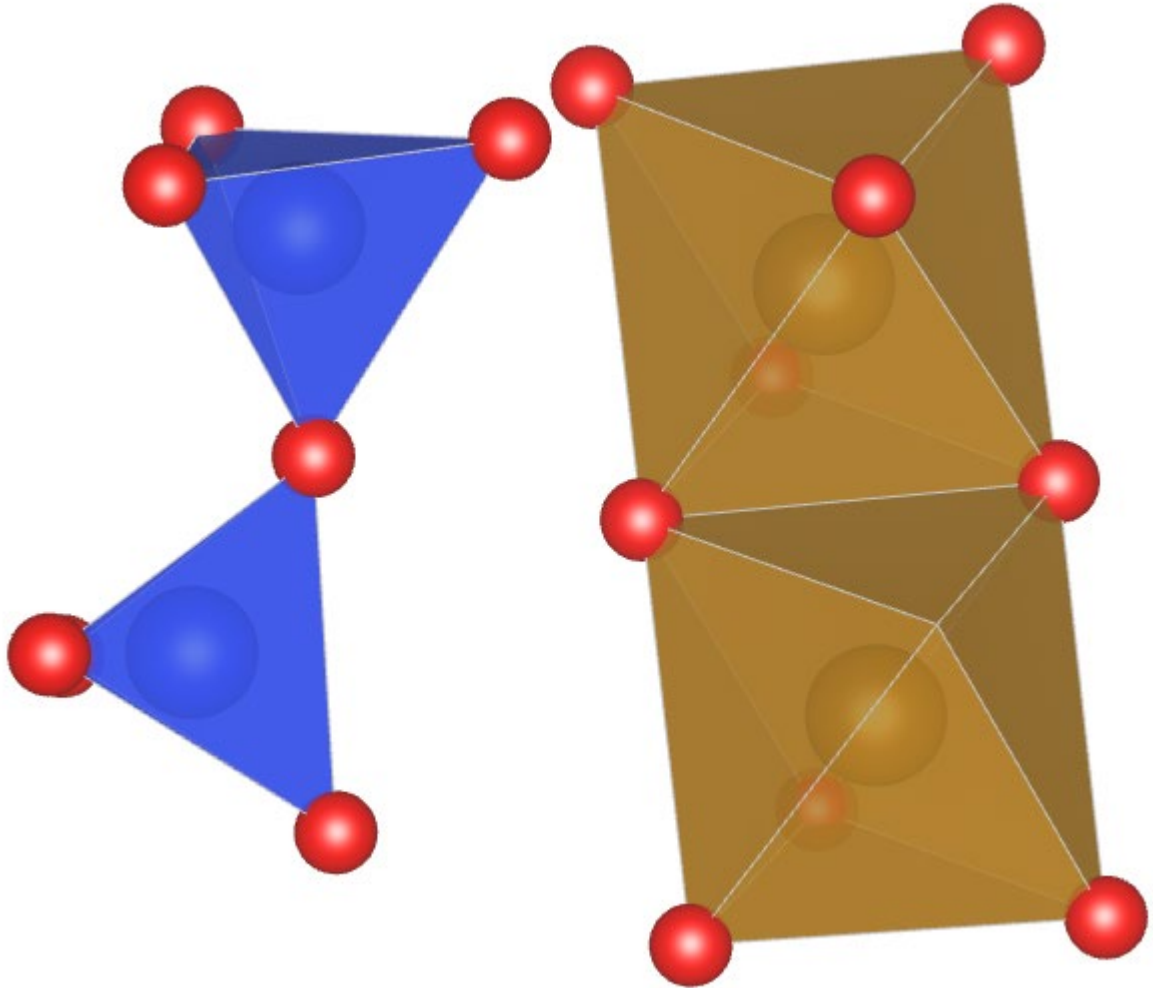
Crystal structure has different features at different scales



Local environment

- Coordination number
- Angles
- Distortions (Jahn-Teller etc)

Crystal structure has different features at different scales

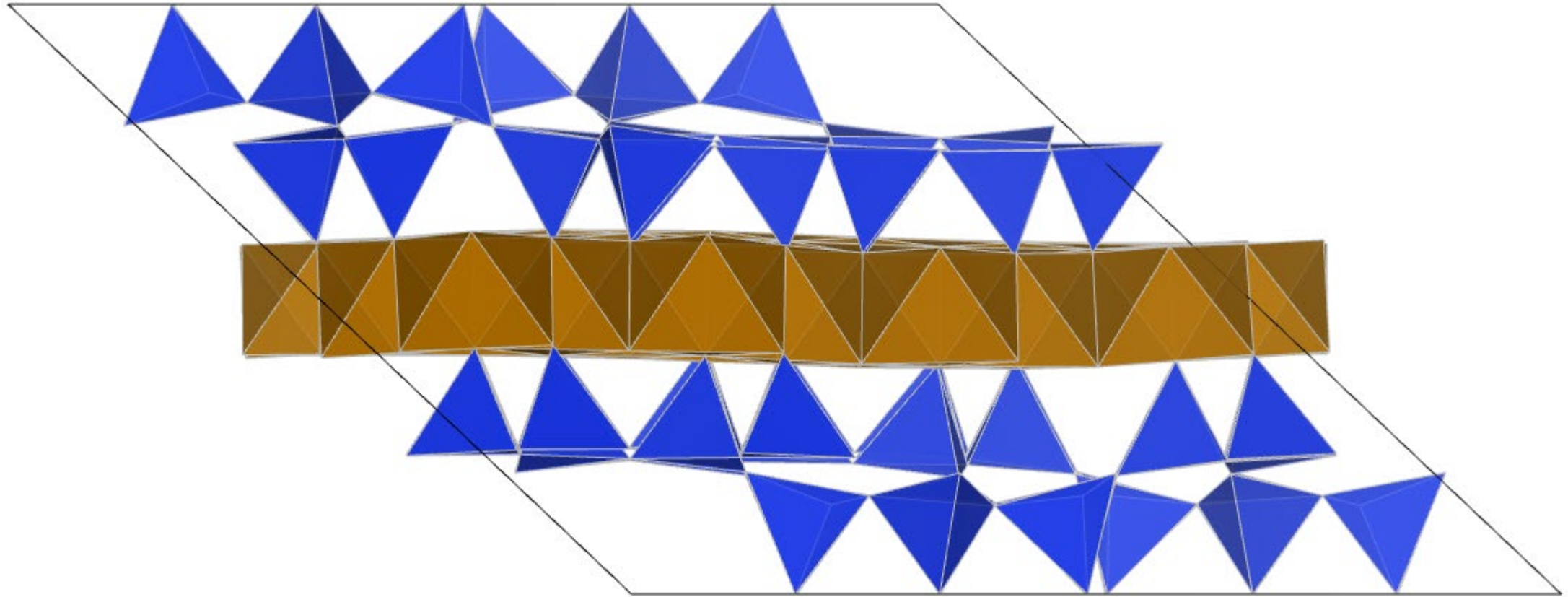


Polyhedra

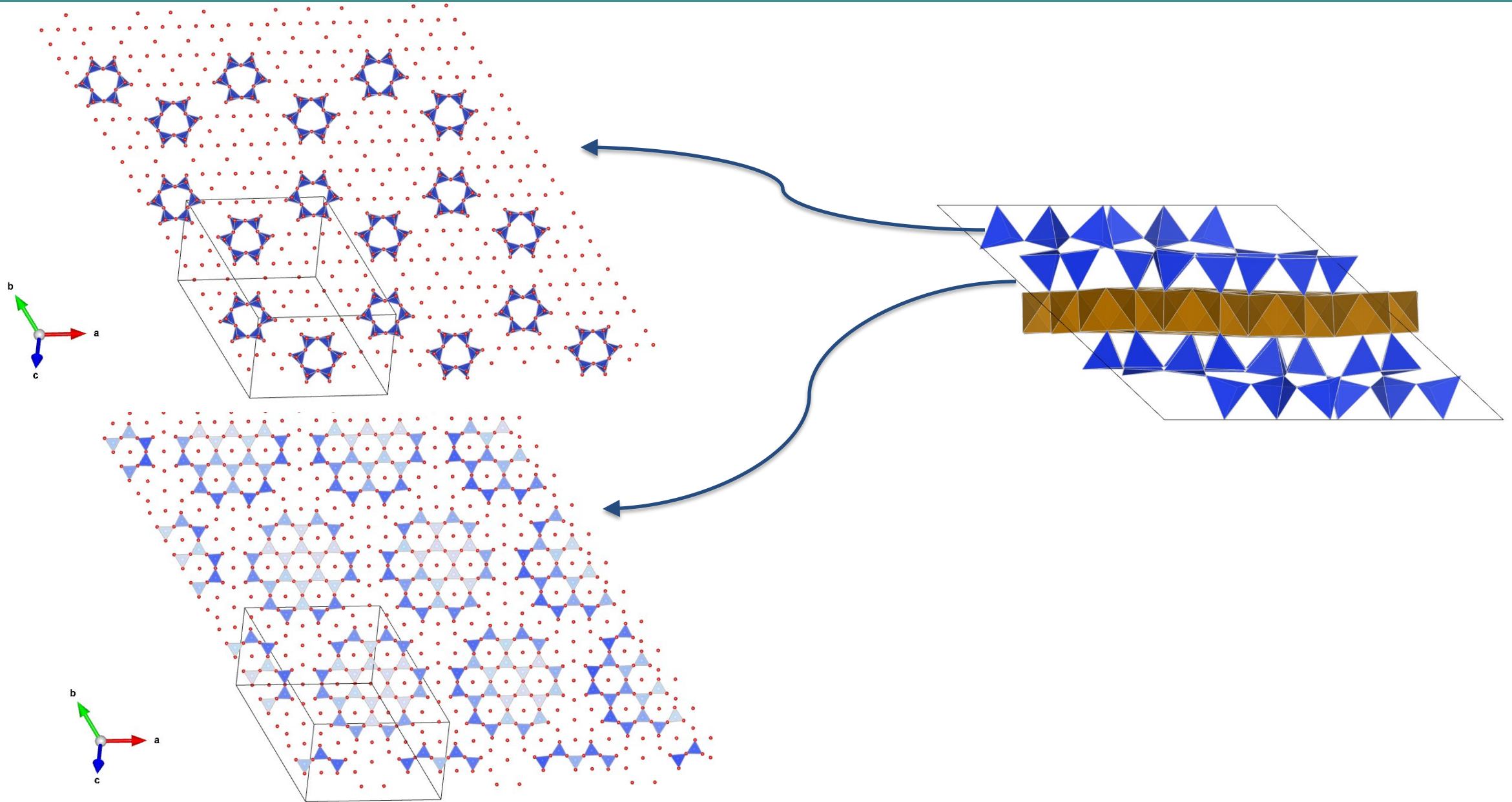
- isolated vs corner vs edge vs face sharing
- multiple types
- angles
- distances



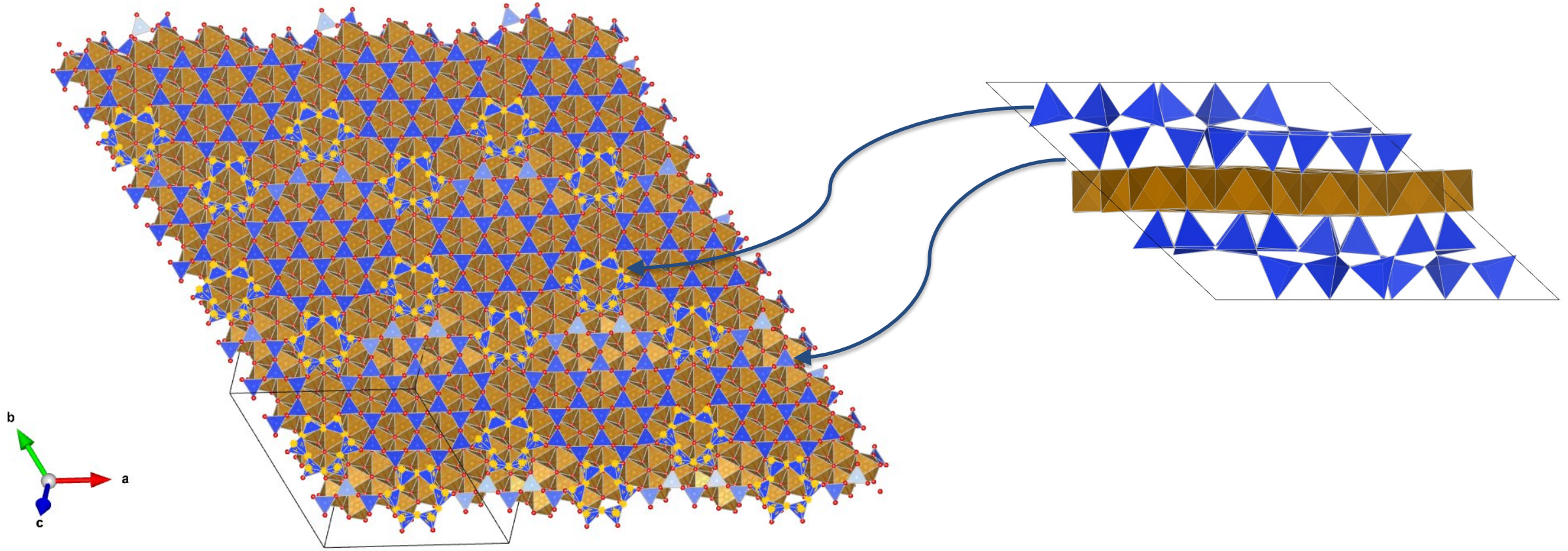
Atomic features can be atomic scale... Or unit cell scale...



Some unit cell features are “slabs” or layers

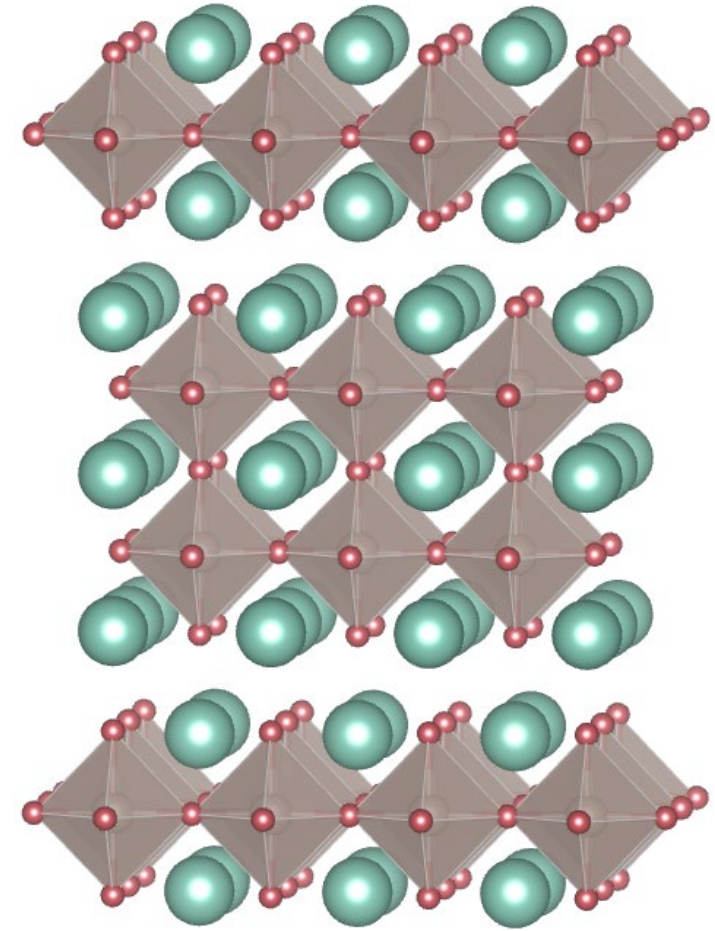
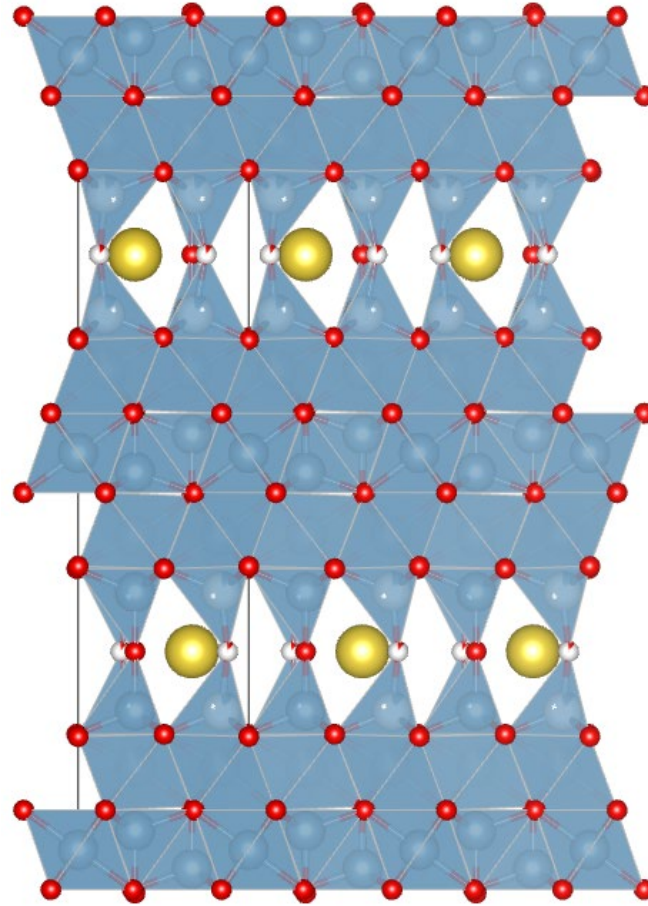
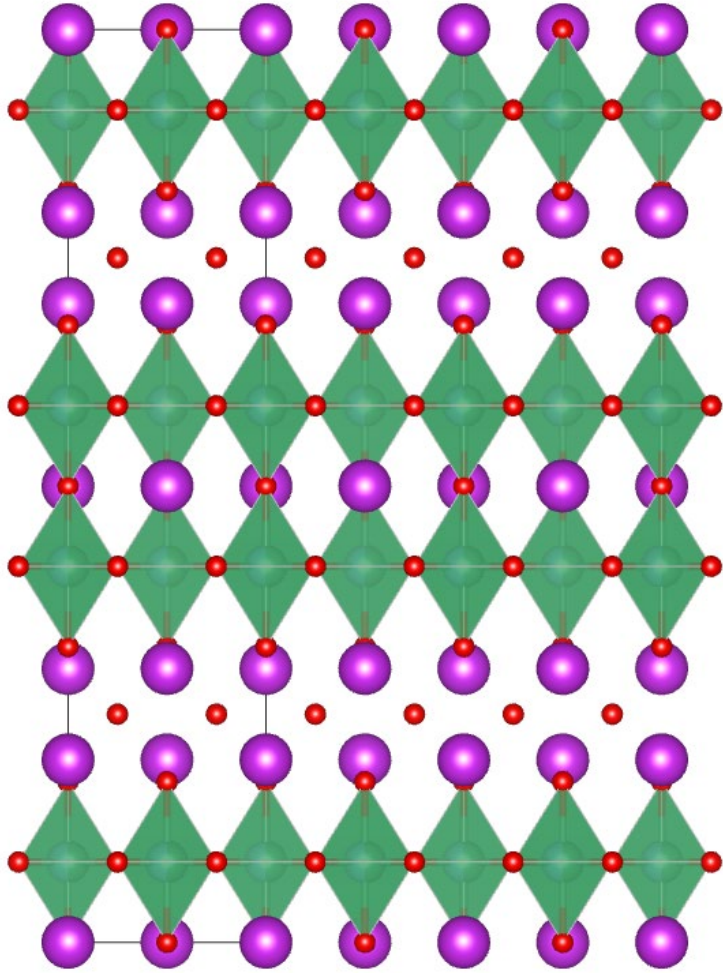


Layers don't need to be continuous



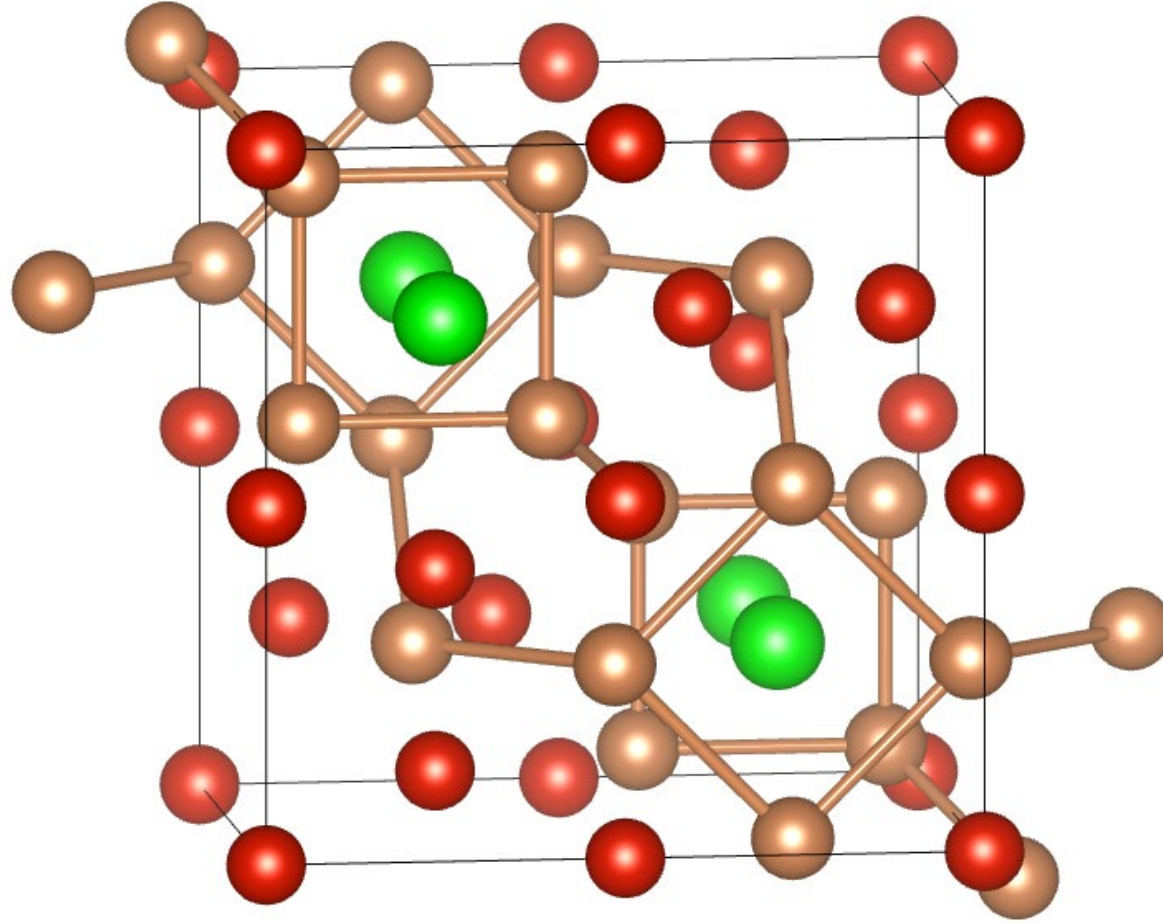


Whole structures are made up of slabs



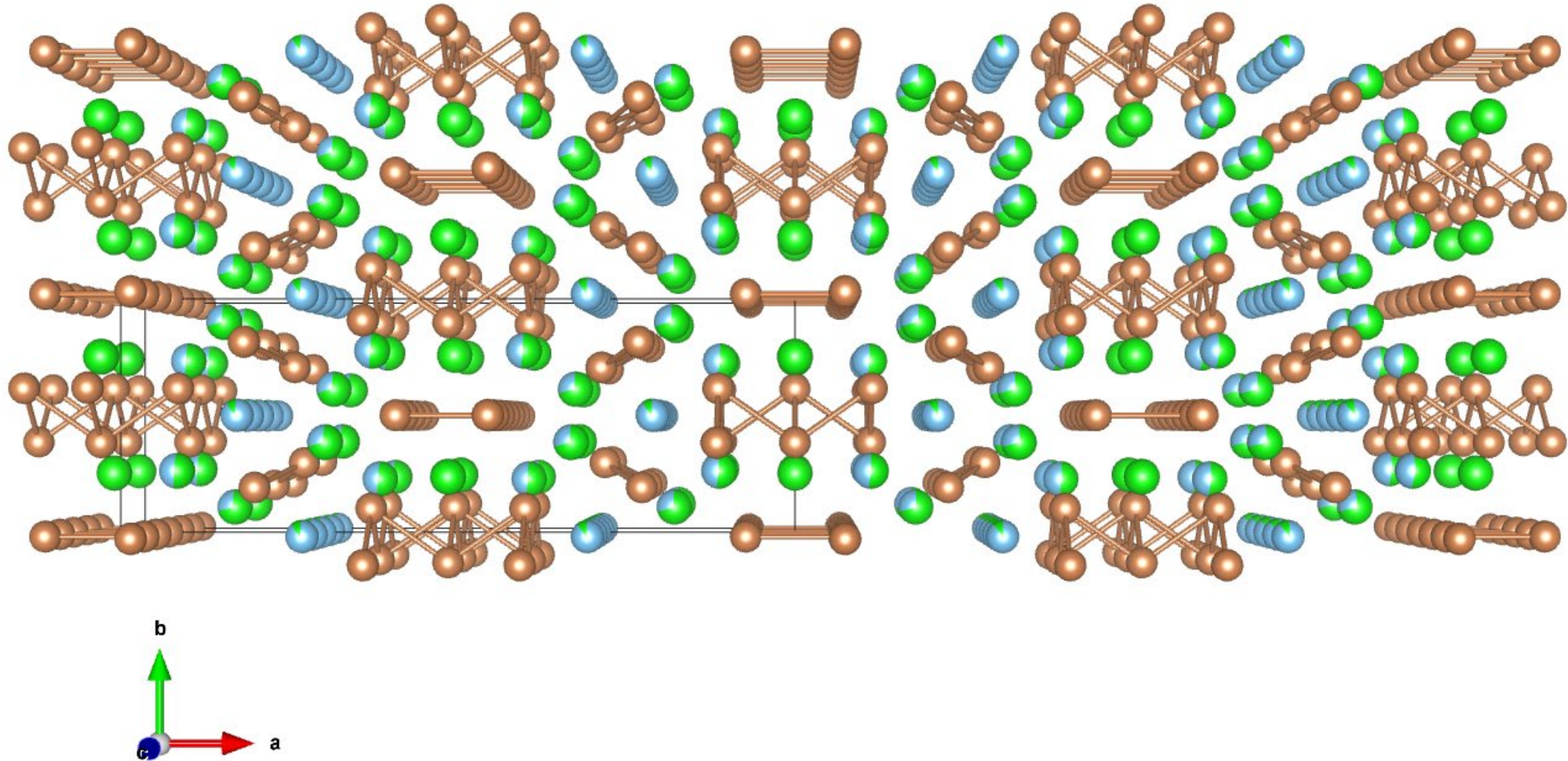


Chains and planes of atoms can define properties



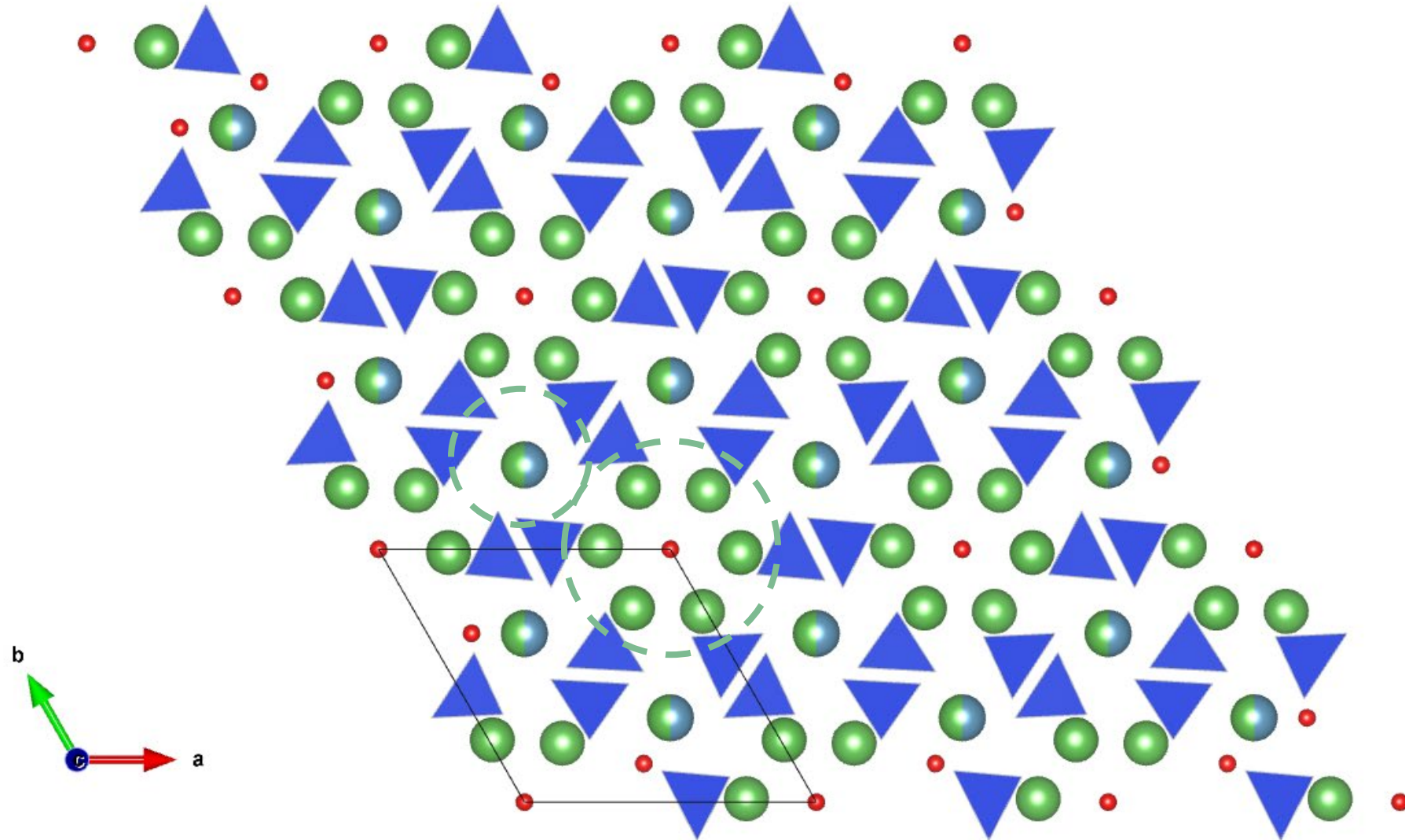


Chains and planes of atoms can define properties



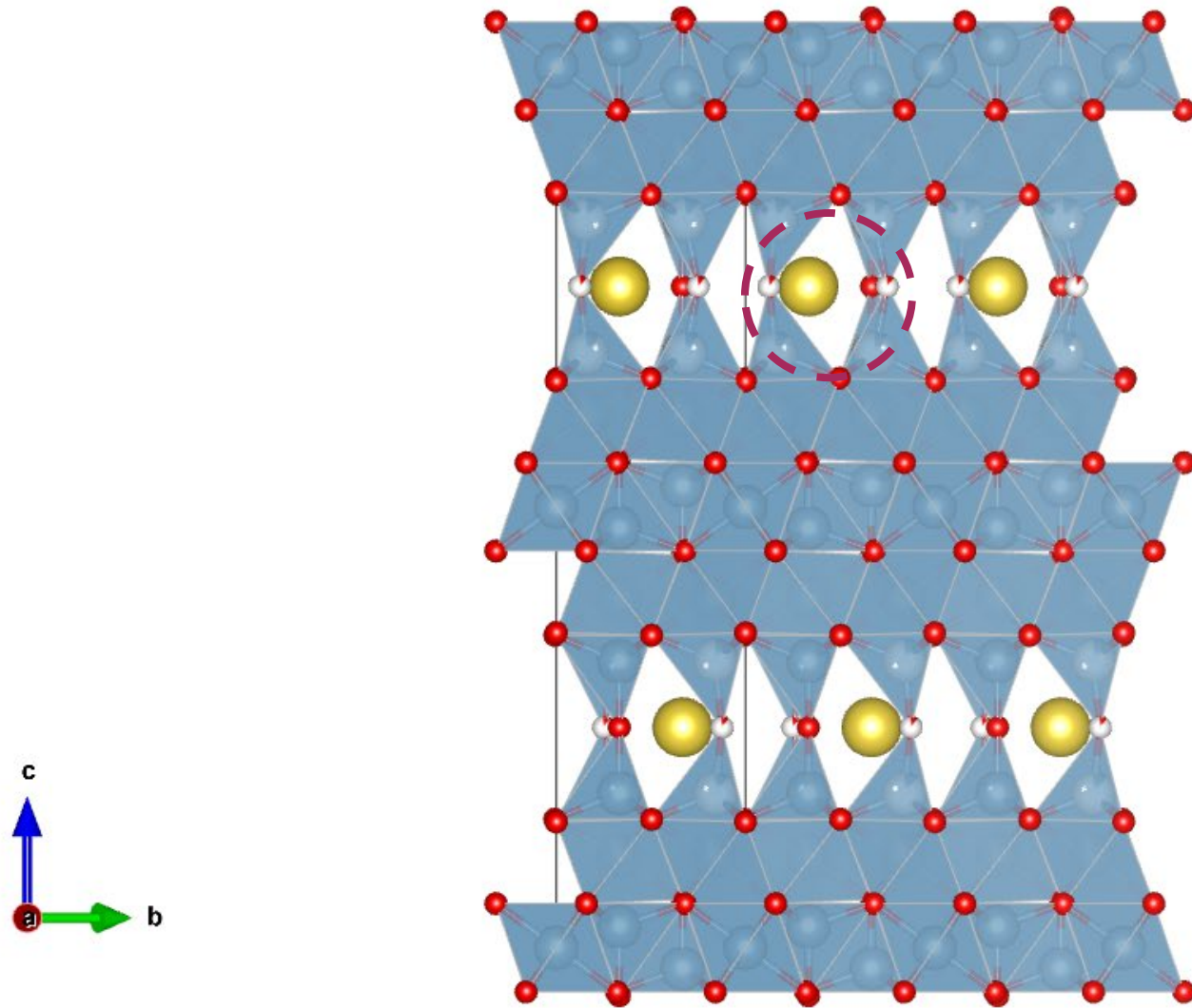


Negative space not occupied by atoms can be the feature



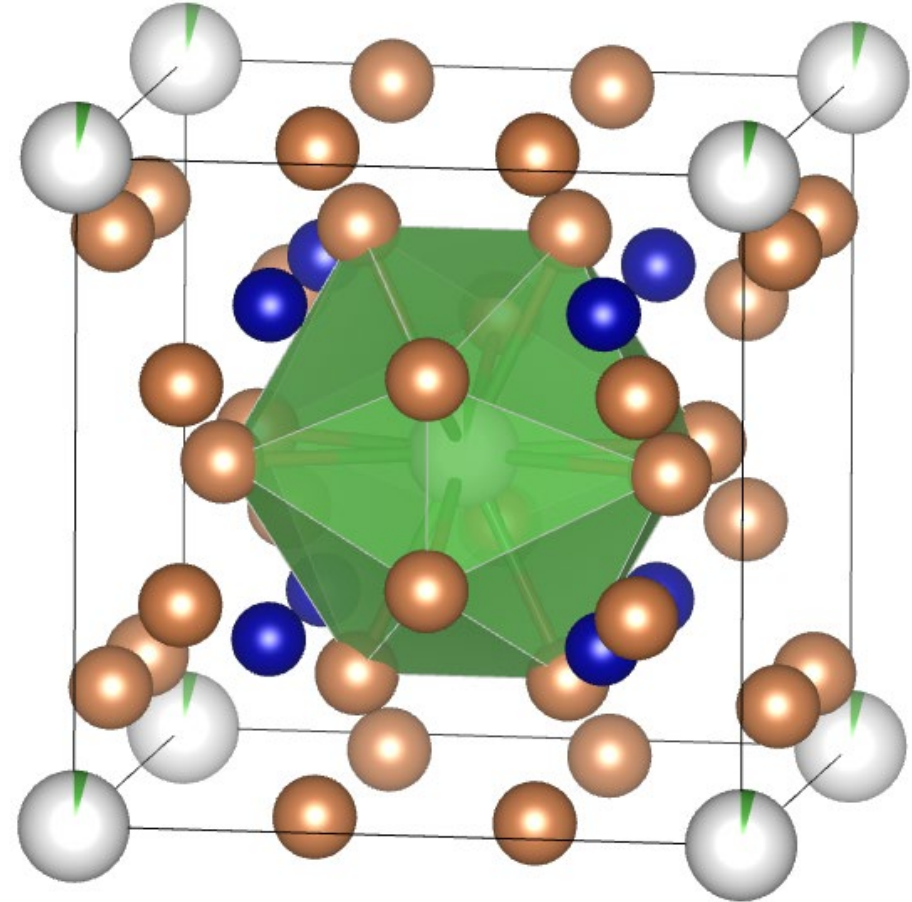
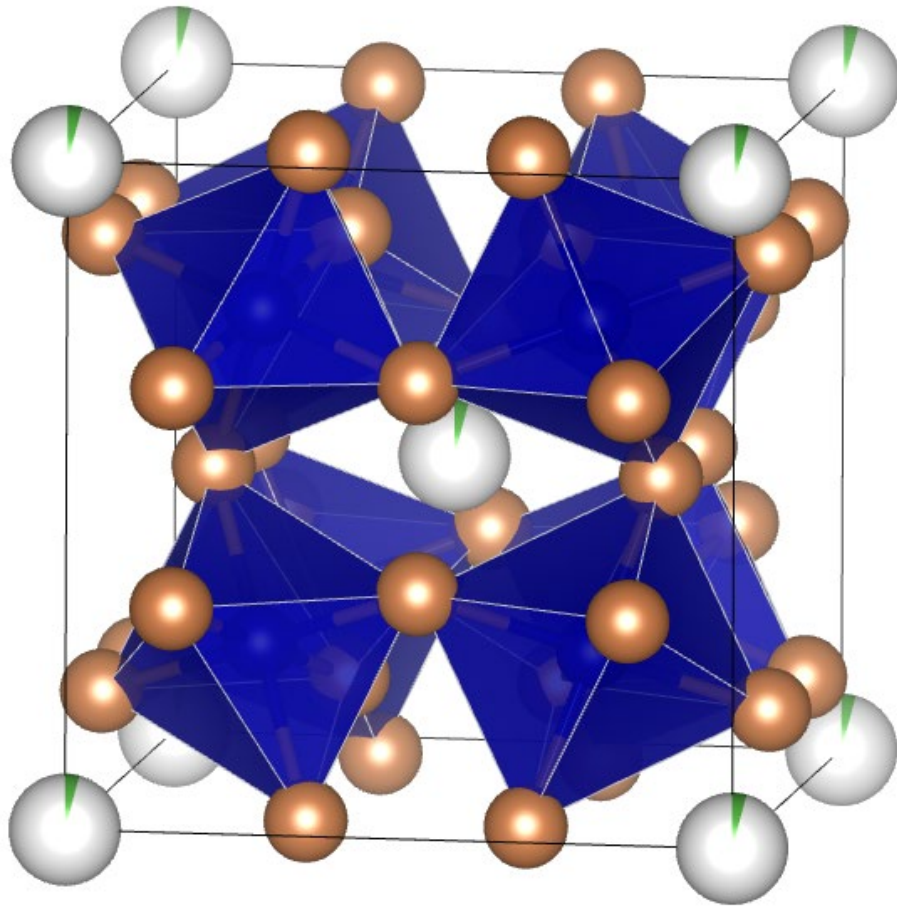


Negative space not occupied by atoms can be the feature

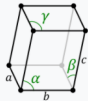
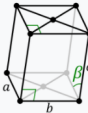
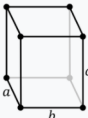
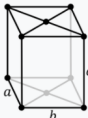
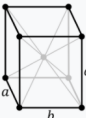
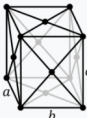
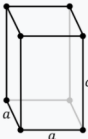
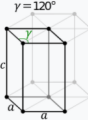
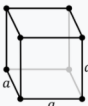
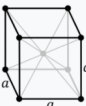
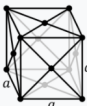




Negative space not occupied by atoms can be the feature



We could define structure with symmetry

Crystal family		Lattice system	Point group (Schönflies notation)	14 Bravais lattices			
				Primitive (P)	Base-centered (S)	Body-centered (I)	Face-centered (F)
Triclinic (a)			C_i				
				aP			
Monoclinic (m)			C_{2h}				
				mP	mS		
Orthorhombic (o)			D_{2h}				
				oP	oS	oI	oF
Tetragonal (t)			D_{4h}				
				tP		tI	
Hexagonal (h)	Rhombohedral		D_{3d}				
				hR			
	Hexagonal		D_{6h}				
				hP			
Cubic (c)			O_h				
				cP		cI	cF

7 crystal systems

14 bravais lattices

Lattice parameters (a,b,c)

Angles (α, β, γ)

Centering (primitive, base, body, face)

32 Schoenflies point groups

We could define structure with symmetry

230 The Space Group List Project



230 space groups for unique symmetries
Symmetry is applied to crystal basis

How do we encode all of these features?

Structural feature	Specific structure
Layered	Yes
Channels	No
RS blocks	Yes
Octahedra	Yes
Tetrahedra	No
...	
Corner-sharing	Yes
Edge-sharing	No

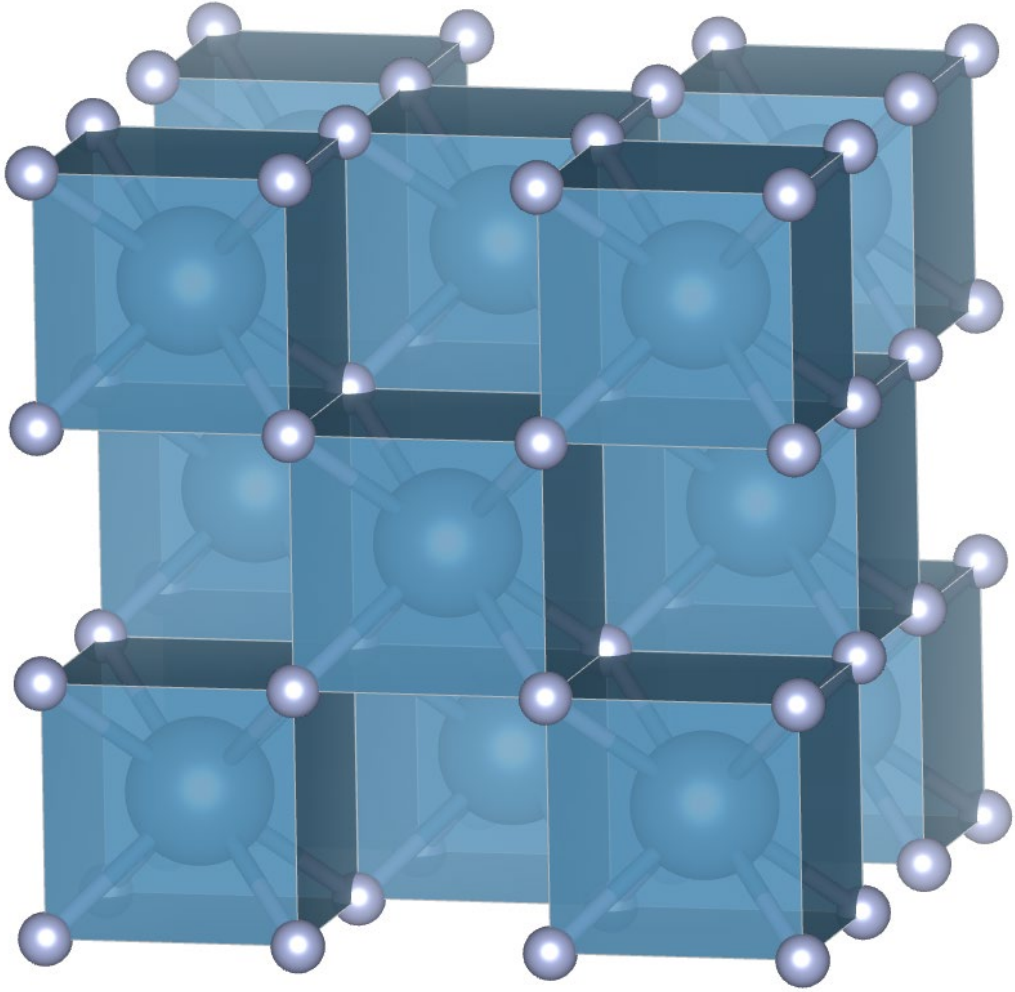
Maybe we could one-hot encode them?

How do we encode all of these features?

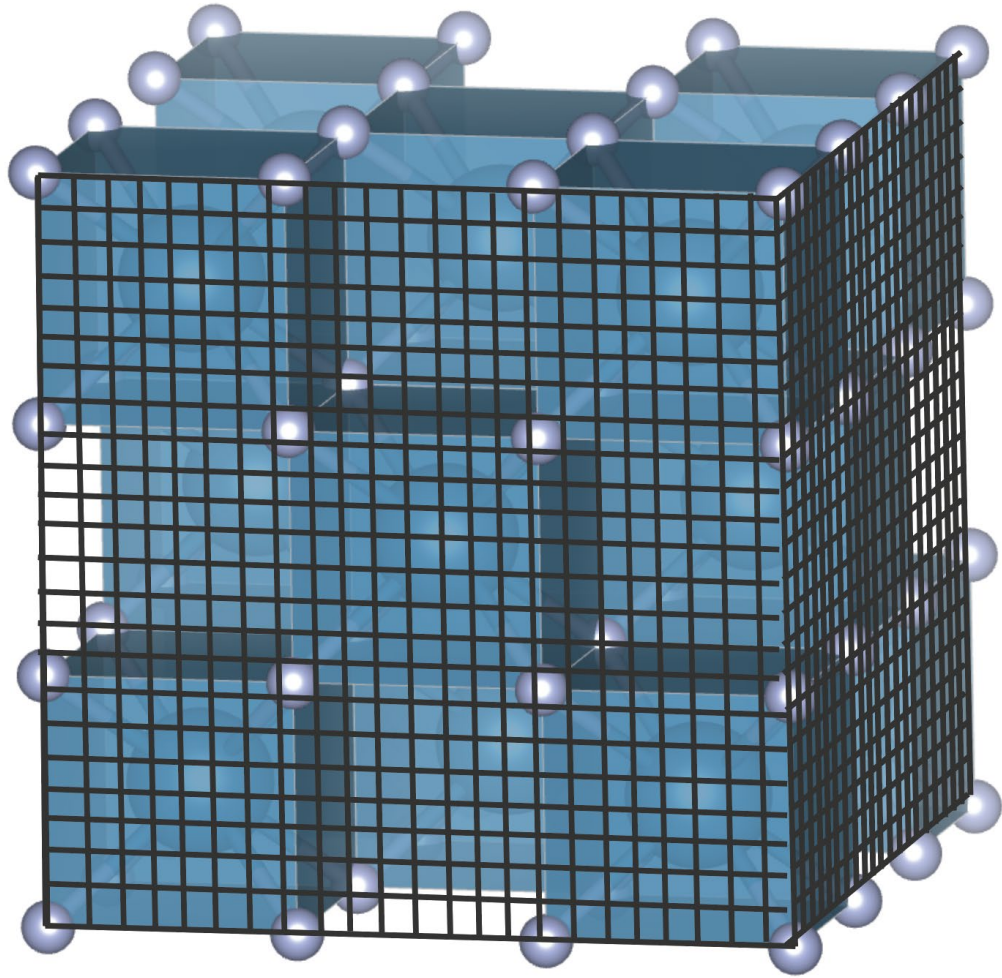
Structural feature	Ruddlesden-Popper	Aurivillius	Spinel	Rock Salt
Layered	1	1	0	0
Channels	0	0	1	0
RS blocks	1	0	0	1
Octahedra	1	1	1	1
Tetrahedra	0	0	1	0
...				
Corner-sharing	1	1	1	1
Edge-sharing	0	0	1	0

Maybe we could one-hot encode them?

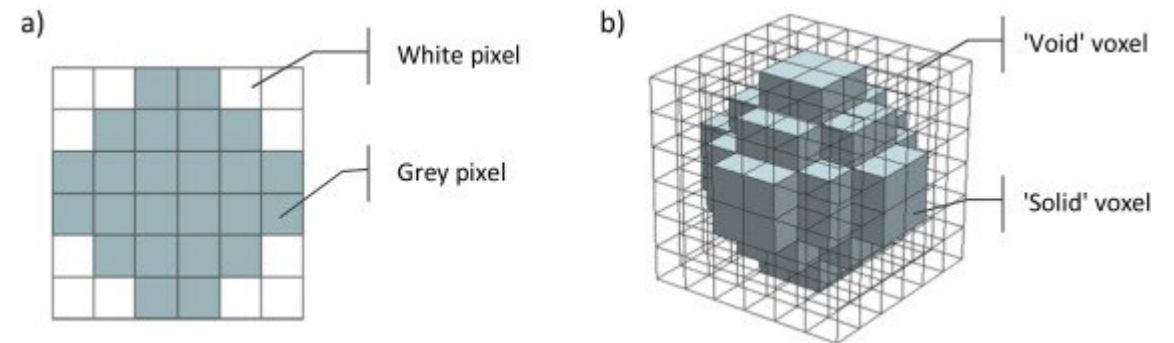
We could chop the crystal up into tiny voxels occupied or not



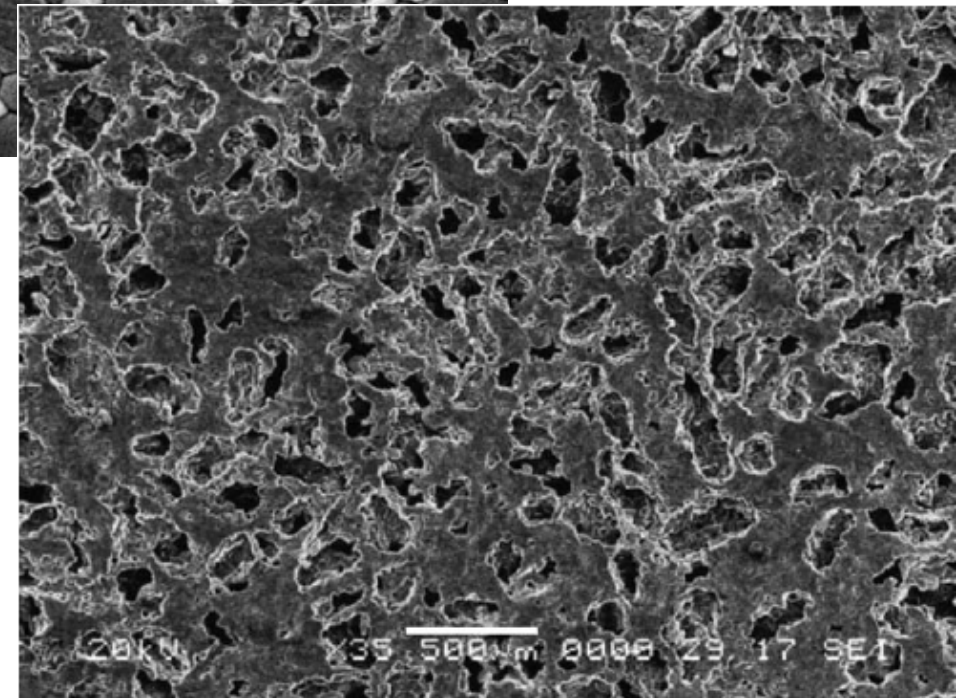
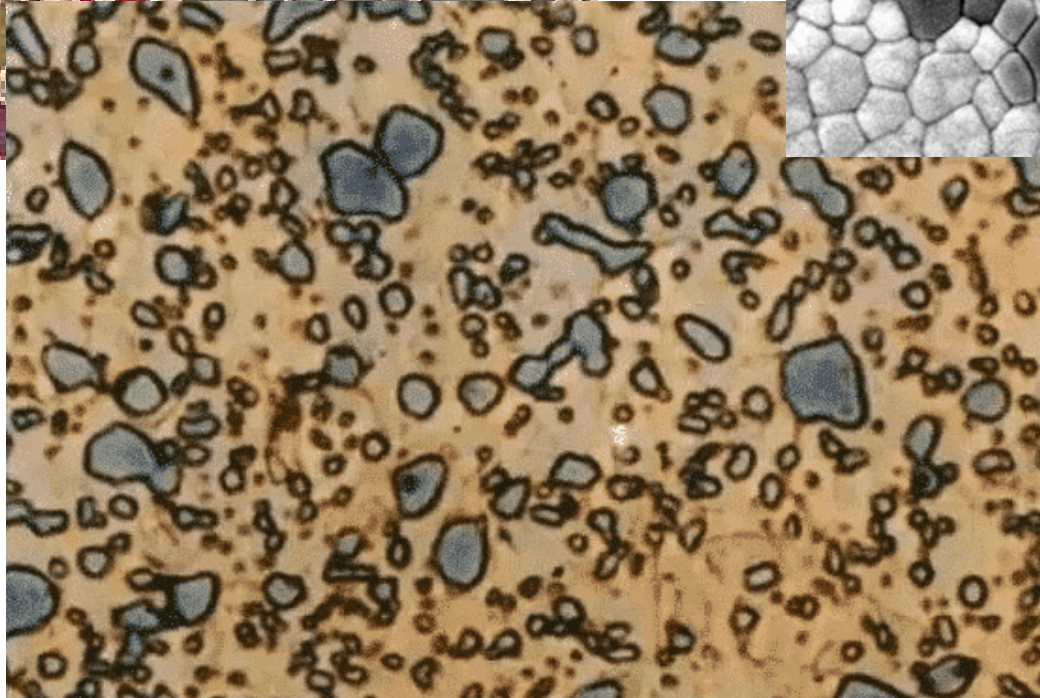
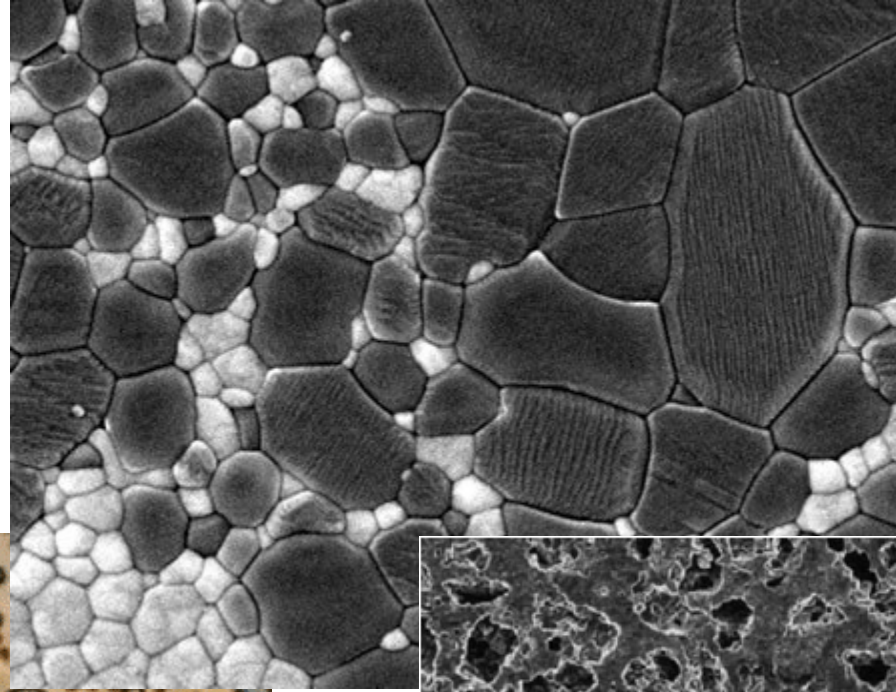
We could chop the crystal up into tiny voxels occupied or not



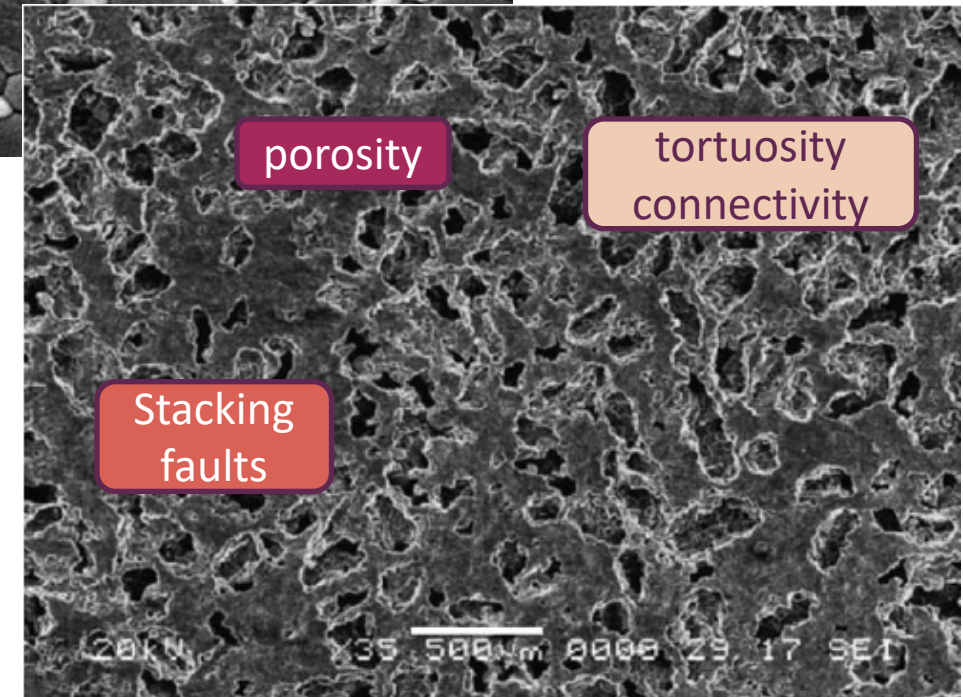
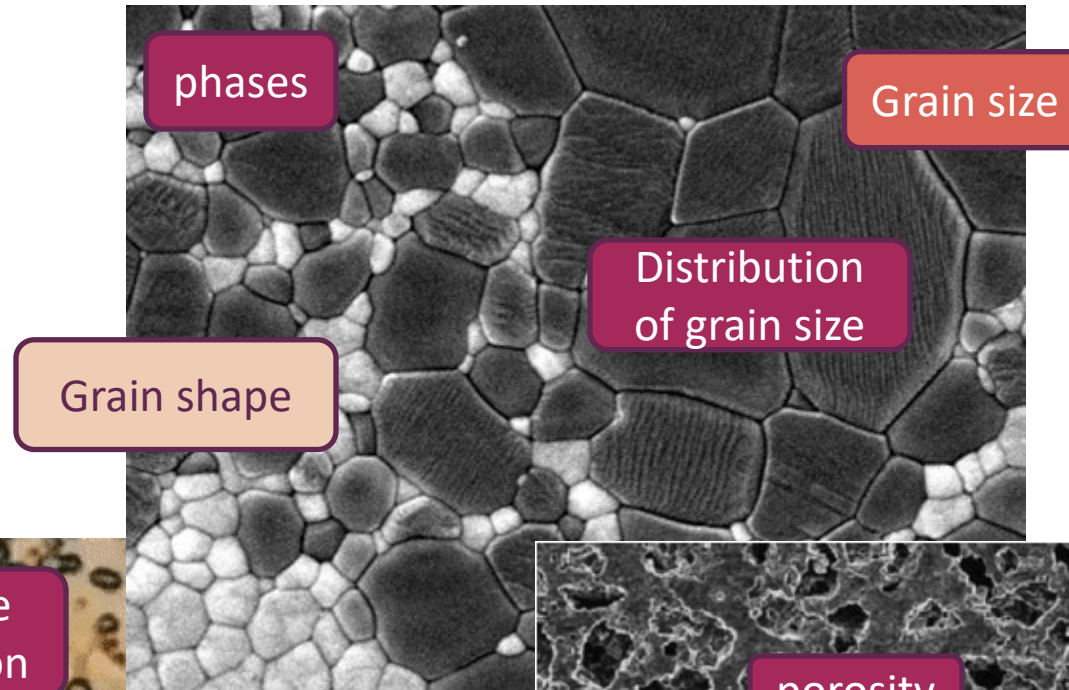
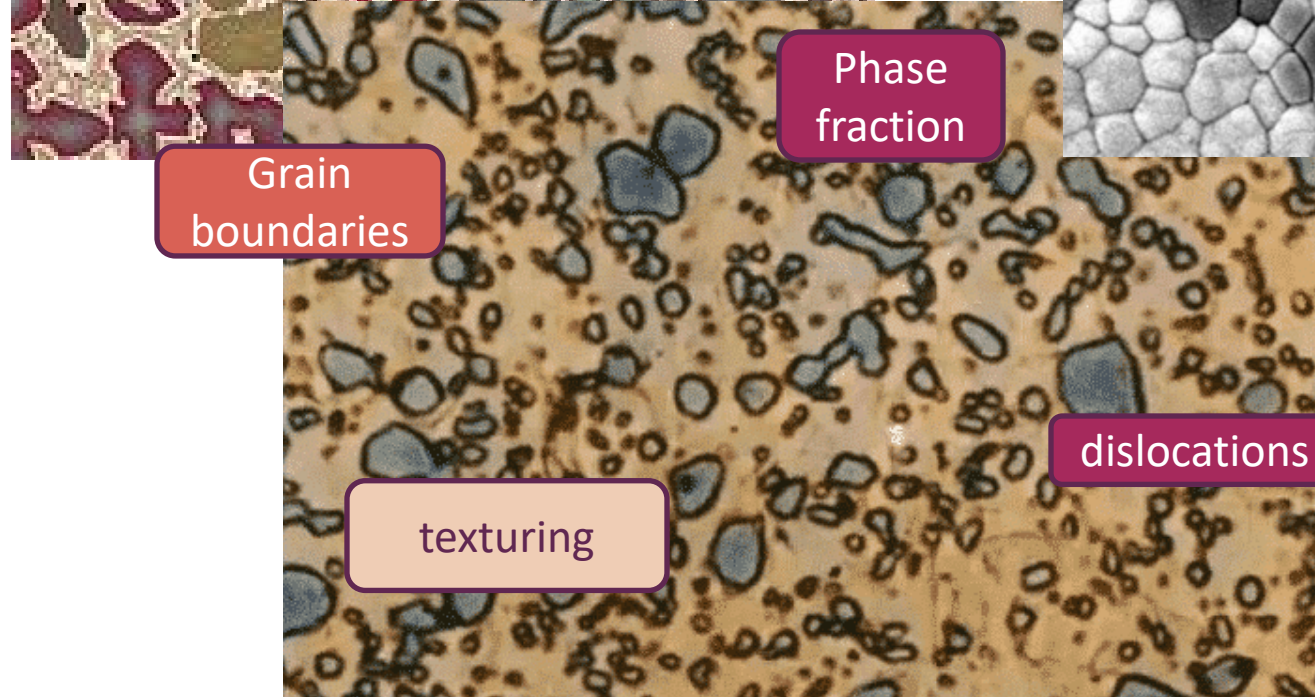
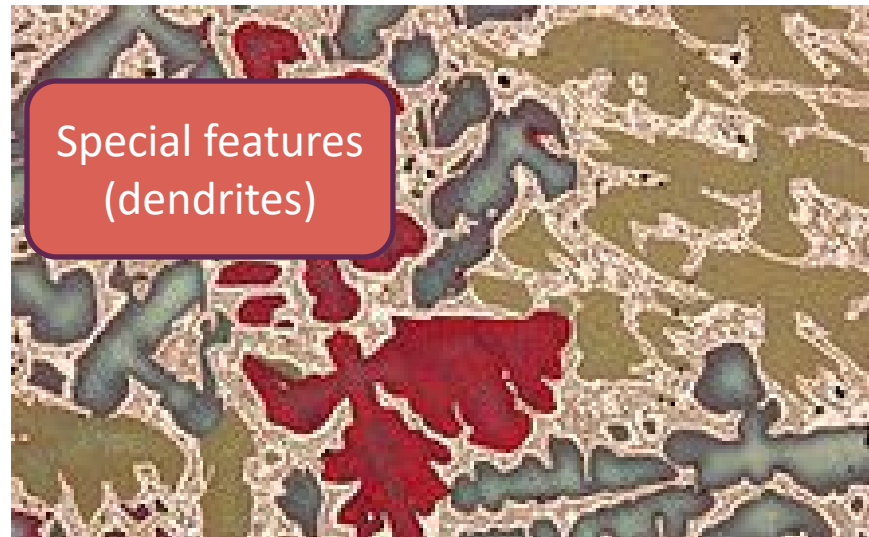
Do put a point at atom center, or do we try to account for atom size?
How do we encode different atom chemistries?



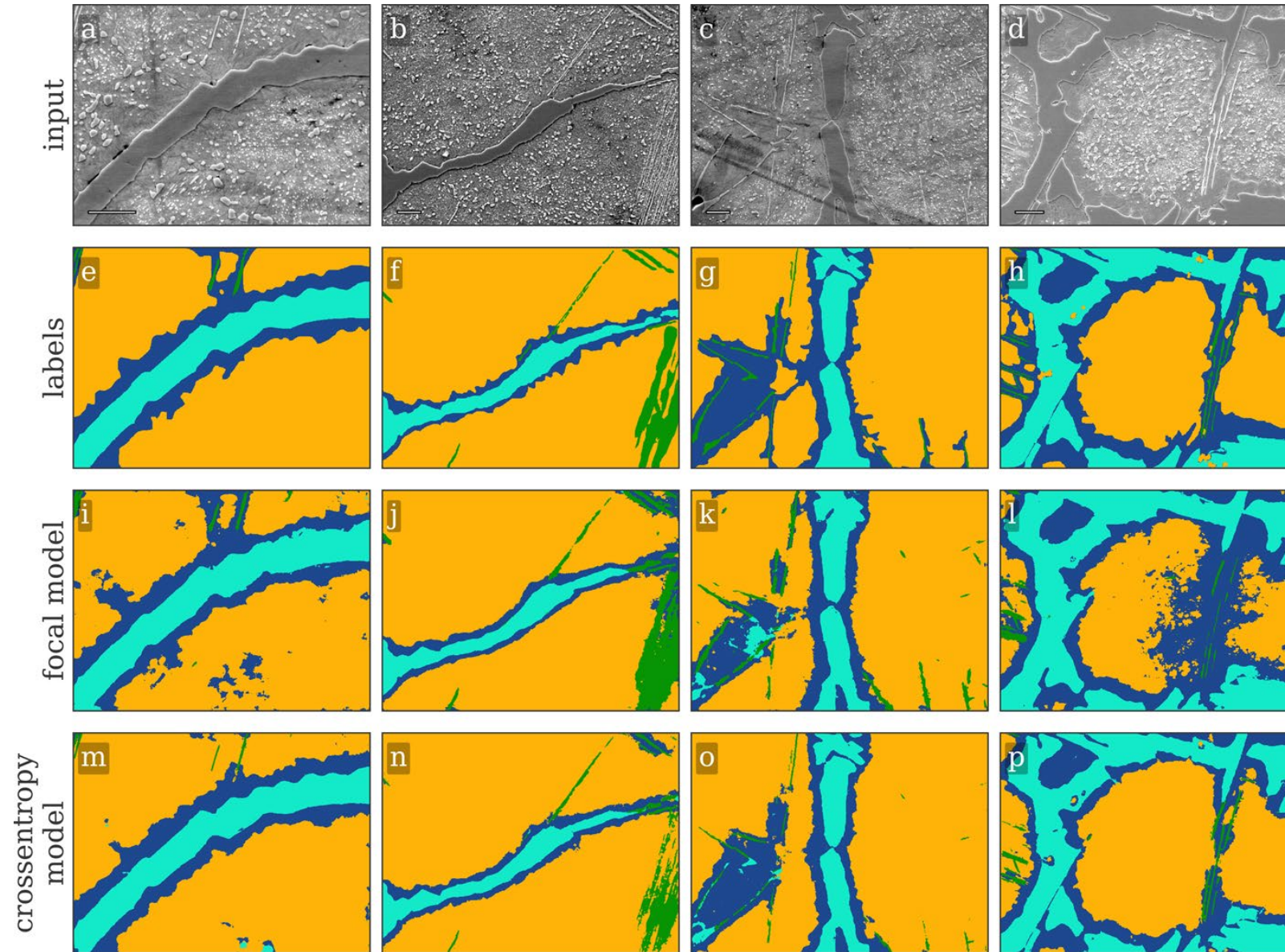
What information is present at the microstructure scale?



What information is present at the microstructure scale?



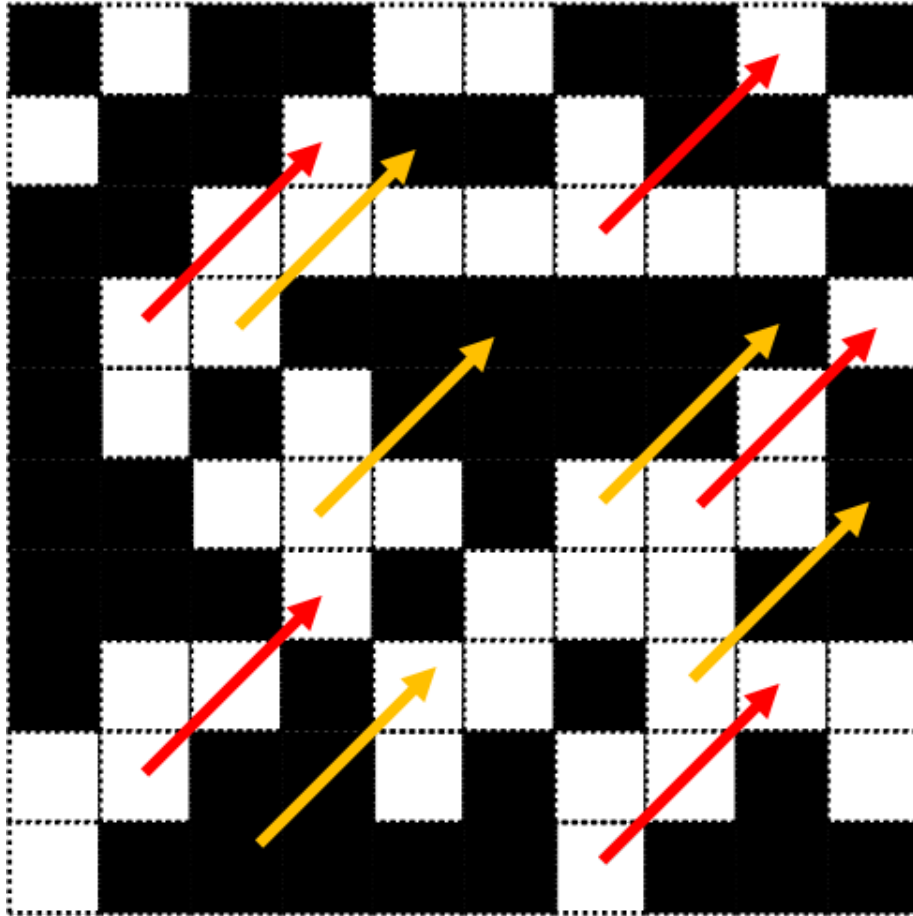
How do we identify and featurize these features?



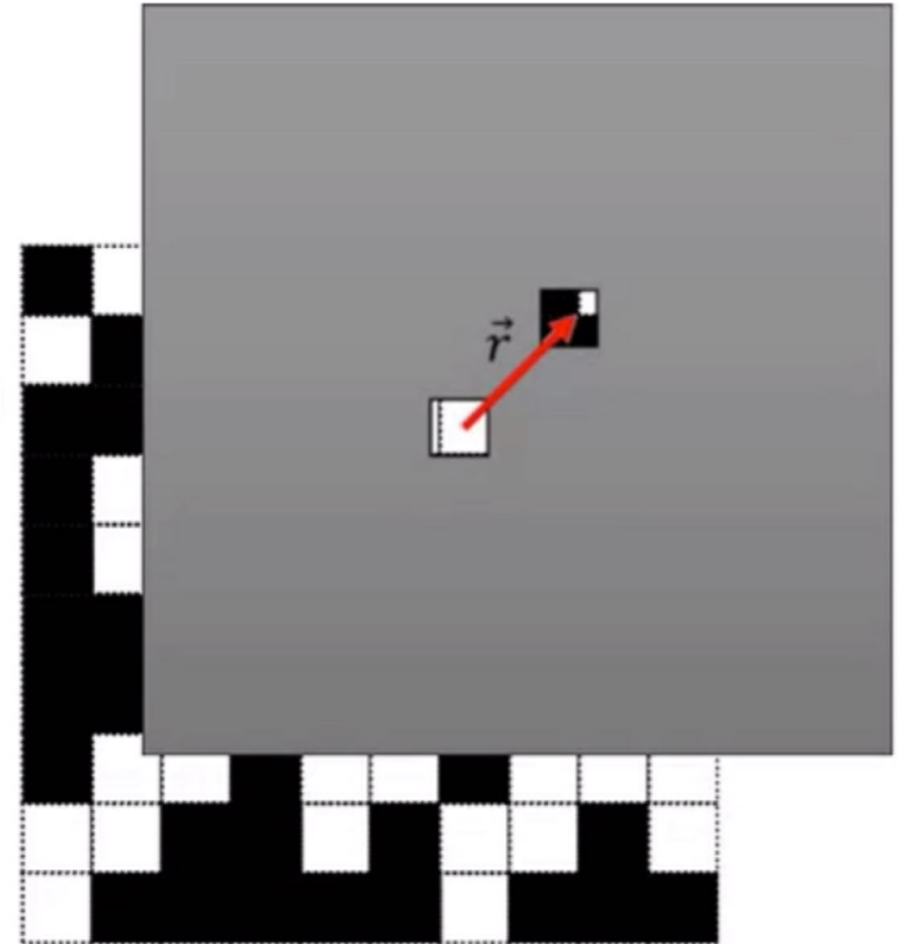


2-pt statistics is one powerful tool for quantifying these features

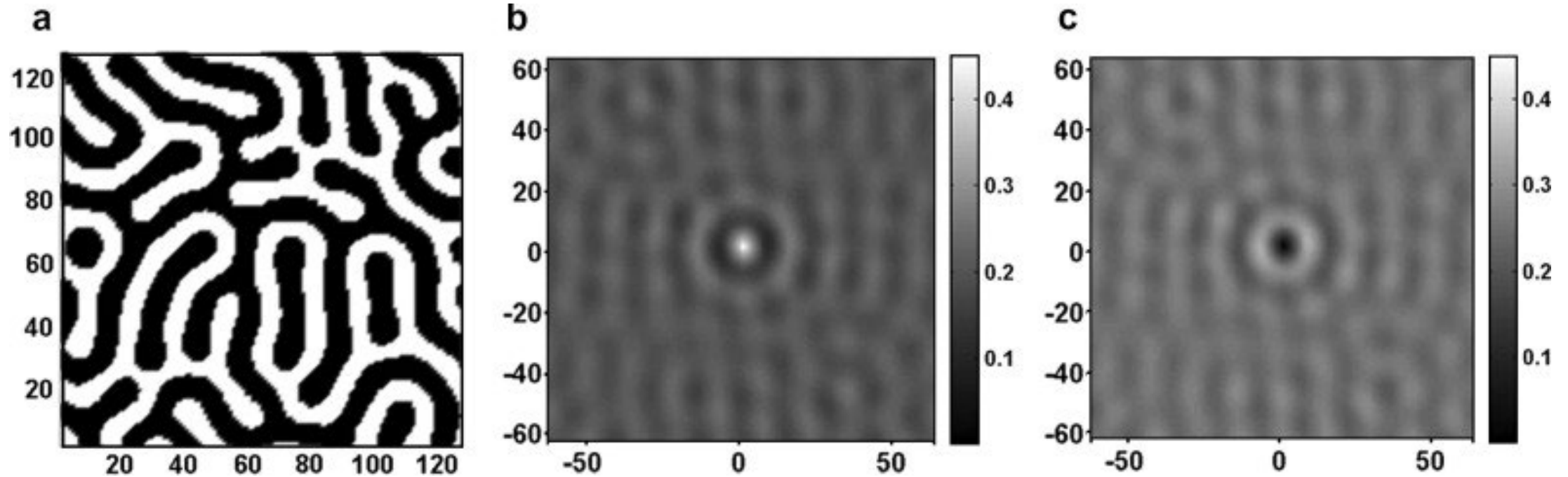
$$f_r^{hh'} = \frac{\# \text{ Trials Successful}}{\# \text{ Trials Attempted}}$$



$$P(A^h \cap A_{@ \vec{r}}^{h'} | \vec{r}) = \frac{P(A^h \cap A_{@ \vec{r}}^{h'} \cap \vec{r})}{P(\vec{r})}$$



2-pt statistics is one powerful tool for quantifying these features

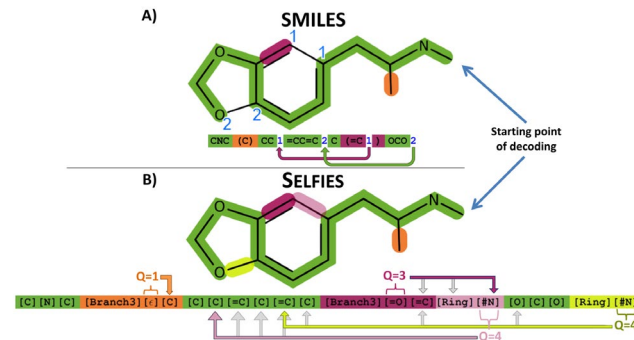


Fullwood, David T., Stephen R. Niezgoda, and Surya R. Kalidindi. Acta Materialia 56.5 (2008): 942-948.

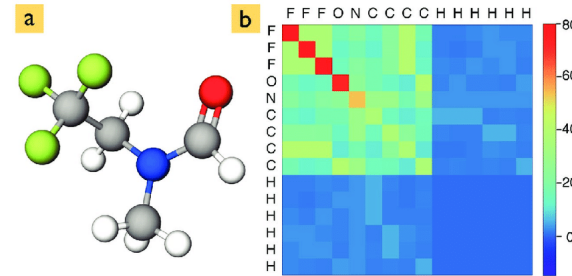
Microstructure
segmentation/analysis



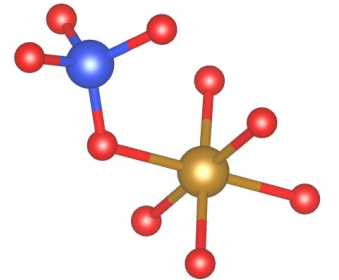
String representations of molecules



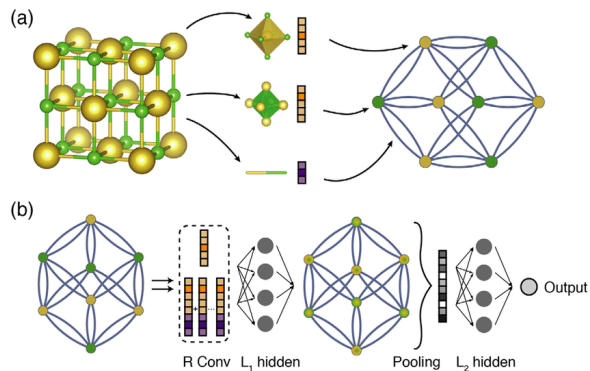
Coulomb matrix



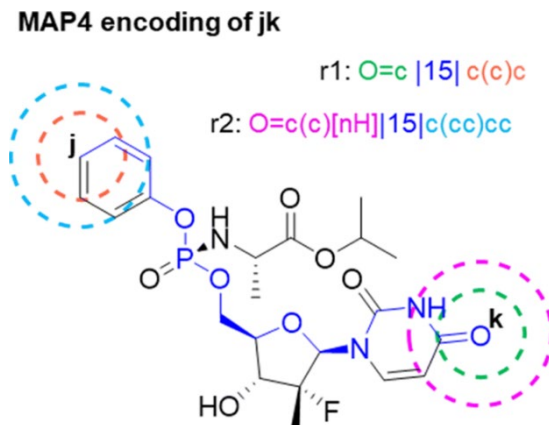
3D coordinates



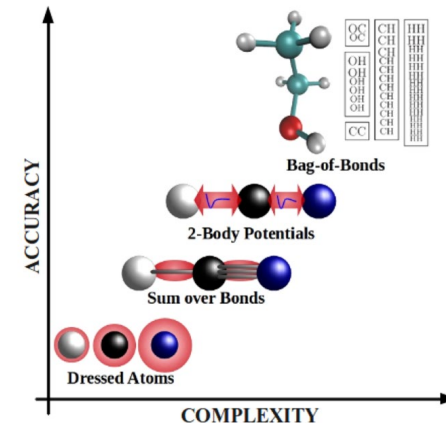
Crystal graphs



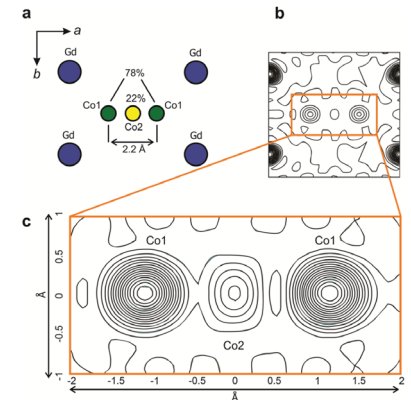
Fingerprints



Bag of Bonds /
Fragments



Electron density



Crystal Structure graphs

